

GRENONBLE LE VERSOUD AP, France

WMO: 074870

Lat: 45.2173N

Lon: 5.8487E

Elev: 220

StdP: 98.71

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	-4.6	-9.9	1.7	-4.1	-7.5	2.1	-2.1	8.1	11.7	6.9	7.0	1.4	30	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	32.9	22.1	31.1	21.3	29.5	20.6	23.0	30.9	22.0	29.6	21.2	28.1	2.3	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	15.4	26.4	19.5	14.6	25.3	18.7	13.9	24.3	69.0	31.0	65.3	29.8	62.2	28.1	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.1	4.9	-8.4	35.2	3.7	1.6	-11.1	36.4	-13.3	37.3	-15.4	38.3	-18.1	39.5		
			-8.9	25.3	3.6	1.1	-11.5	26.1	-13.6	26.7	-15.6	27.3	-18.2	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	2.9	4.1	8.7	13.0	16.1	20.4	21.9	21.1	17.7	13.4	7.3	3.2
	DBStd	7.61	3.45	3.70	3.40	3.38	3.53	3.21	3.08	3.03	2.97	3.46	3.80	3.36
	HDD10.0	787	222	166	66	11	1	0	0	0	0	12	97	213
	HDD18.3	2462	479	399	300	161	88	15	7	10	46	156	331	470
	CDD10.0	1707	1	1	24	102	189	313	369	344	231	116	16	2
	CDD18.3	339	0	0	0	1	18	78	118	96	27	2	0	0
	CDH23.3	3363	0	0	1	60	224	764	1163	892	236	23	0	0
	CDH26.7	1178	0	0	0	7	53	269	458	338	51	1	0	0
Wind	WSAvg	1.4	1.2	1.5	1.6	1.7	1.6	1.6	1.5	1.3	1.2	1.2	1.2	1.2
Precipitation	PrecAvg	1208	97	82	82	96	110	97	92	95	111	118	127	100
	PrecMax	1497	267	248	285	217	210	171	220	167	278	304	328	237
	PrecMin	909	16	10	17	28	43	35	29	20	20	19	31	2
	PrecStd	172	56	56	54	46	44	38	45	34	64	59	74	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	17.5	22.6	27.8	30.6	34.1	34.9	34.5	30.4	25.9	20.5	16.1
		MCWB	10.1	10.0	12.7	16.1	20.3	22.6	22.3	22.2	20.1	17.5	12.4	10.3
	2%	DB	11.8	14.6	20.2	25.0	28.1	31.7	33.0	32.4	28.3	23.3	17.9	12.0
		MCWB	8.5	8.6	11.8	15.3	18.4	21.5	22.1	22.0	19.8	16.5	12.1	9.0
	5%	DB	9.2	12.3	18.0	22.9	25.9	29.8	31.2	30.4	26.1	21.3	15.1	9.7
		MCWB	7.0	7.5	11.1	14.2	17.3	20.7	21.5	21.3	18.4	15.7	11.3	7.5
	10%	DB	7.5	9.9	15.6	20.8	23.5	27.9	29.3	28.3	24.2	19.2	13.0	7.9
		MCWB	5.9	6.6	10.0	13.3	16.3	19.8	20.7	20.5	17.8	14.9	10.6	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.6	13.6	16.7	20.8	24.2	24.2	23.9	21.9	18.4	14.1	11.0
		MCDB	15.0	15.9	20.2	25.3	29.2	31.7	32.7	31.9	27.7	23.4	17.0	14.3
	2%	WB	8.7	9.1	12.4	15.7	19.2	22.5	22.9	22.7	20.2	17.2	12.9	9.5
		MCDB	11.0	13.4	18.6	23.9	26.7	29.9	31.1	30.5	25.9	21.5	16.0	11.7
	5%	WB	7.3	8.0	11.4	14.7	18.1	21.3	22.0	21.7	19.2	16.3	11.8	7.8
		MCDB	8.8	11.4	16.9	21.8	24.3	28.2	30.0	28.8	24.3	20.1	14.7	9.5
	10%	WB	6.1	6.8	10.4	13.7	16.8	20.3	21.1	20.8	18.3	15.3	10.7	6.6
		MCDB	7.4	9.6	14.9	19.6	22.1	26.6	28.3	27.1	23.0	18.6	13.3	7.8
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.9	11.6	12.5	11.7	12.3	12.6	12.5	11.7	10.7	8.6	7.4
		MCDBR	9.8	13.8	16.4	17.1	15.8	15.4	15.6	16.1	14.8	13.7	11.9	10.1
		MCWBR	6.7	8.4	8.8	8.0	7.0	6.6	6.6	6.9	6.9	7.3	7.2	7.1
	5% WB	MCDBR	9.4	12.5	14.6	15.9	14.4	14.3	14.7	14.9	13.4	12.3	10.5	9.4
		MCWBR	6.6	7.9	7.9	7.7	6.7	6.4	6.4	6.7	6.4	6.7	6.5	6.7
Clear-Sky Solar Irradiance	taub	0.297	0.324	0.368	0.416	0.427	0.426	0.418	0.408	0.391	0.366	0.329	0.297	
	taud	2.414	2.336	2.273	2.182	2.215	2.265	2.290	2.326	2.358	2.419	2.452	2.442	
	Ebn at Noon	816	853	858	843	845	845	847	842	825	795	769	776	
	Edh at Noon	72	94	116	138	138	132	127	119	106	87	70	64	
All-Sky Solar Radiation	RadAvg	1.50	2.38	3.59	4.59	5.34	6.36	6.39	5.54	4.24	2.71	1.59	1.27	
	RadStd	0.17	0.30	0.42	0.53	0.47	0.45	0.48	0.35	0.32	0.29	0.21	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page